STRUCTURAL CALCULATION · 3D FRAME · TRAGWERK EXAMPLES

Office building, 3 × 2 bays, 4 storeys

2026-10-21 · Tragwerk 0.2.0 · EN 1992-1-1:2004 with recommended values · 60 nodes, 48 columns, 68 beams · Tragwerk

MAXIMUM UTILISATION

0.8366

Pass

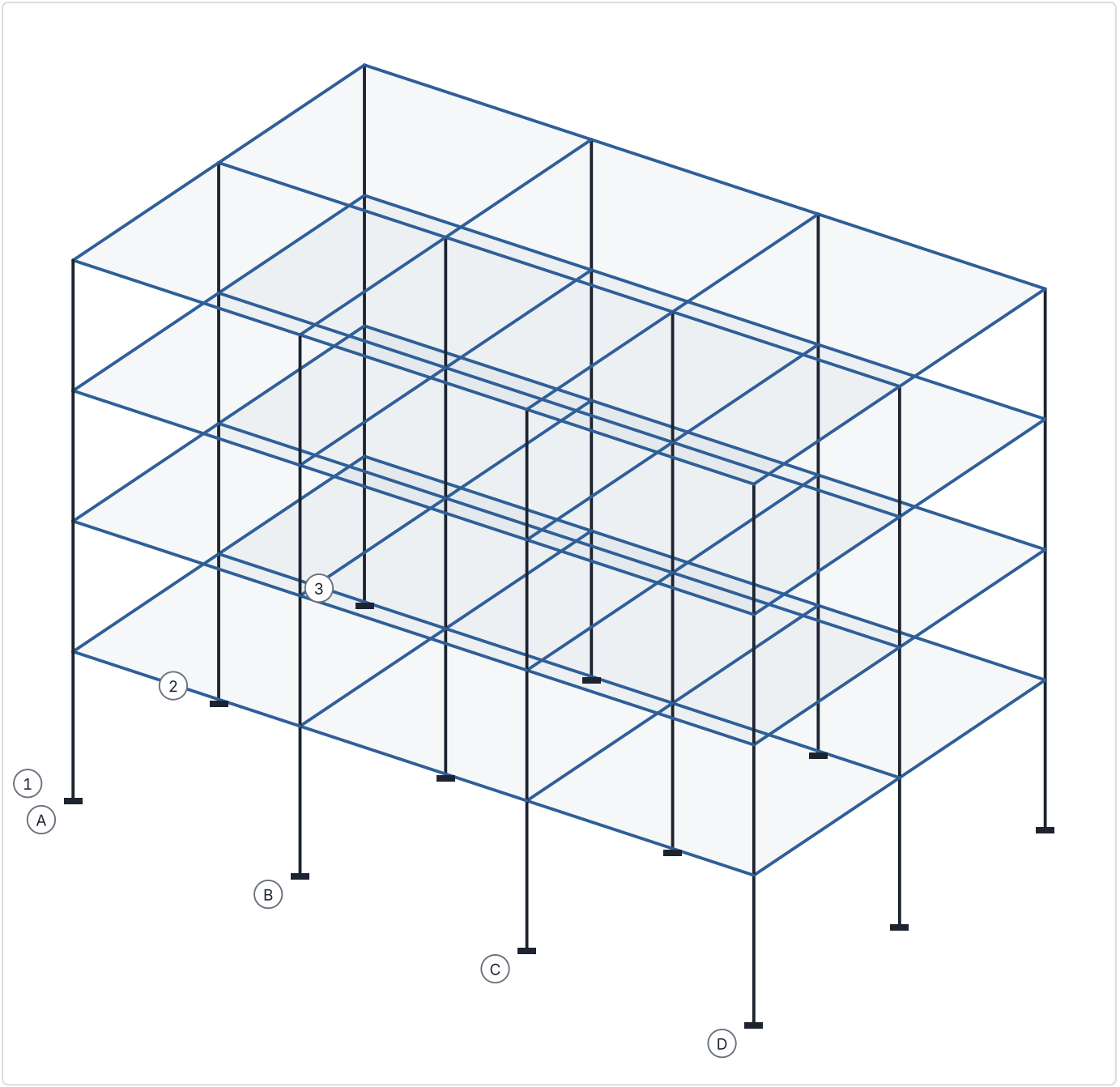
All checks pass

governing: BY-B1-L1, bending with axial force, both axes

1 Basis of design

ITEM	BASIS
Analysis	Linear-elastic 3D frame analysis (direct stiffness method, 6 DOF per node). Members with axial, St Venant torsion and biaxial bending stiffness; gross concrete sections, E = E_{cm}, G = E/(2(1 + ν)), ν = 0.2. Exact member diagrams between nodes.
Slabs	Slabs are not modelled as elements. Their loads reach the beams by tributary areas; no rigid-diaphragm constraint is applied.
Combinations	EN 1990 Eq. (6.10), pattern loading of variable cases, exclusive groups, recommended ψ factors.
Design	EN 1992-1-1:2004 + AC:2010, recommended NDPs: bending with axial force about both axes (6.1, 5.8.9 Eq. 5.39), shear in both planes (6.2), torsion (6.3), slender columns by nominal curvature (5.8.8), detailing (8.2, 9.2, 9.5), deflection (7.4).

2 Structural model



Axonometric view. Dark lines: columns; blue: beams; tinted areas: slab panels (loads only).

GRID	POSITIONS [M]
Axes along X	A = 0, B = 6, C = 12, D = 18
Axes along Y	1 = 0, 2 = 5.5, 3 = 11
Levels Z	L0 = 0, L1 = 3.6, L2 = 6.8, L3 = 10, L4 = 13.2

SECTION	B × H [MM]	CONCRETE [MPA]	LONGITUDINAL BARS	LINKS	MEMBERS
Column	400 × 400	C30/37, f_{cd} = 20.00	top 3Ø20, bottom 3Ø20, each side 1Ø20 (A_s = 2513 mm²)	Ø8 @ 240	48
Beam X	300 × 550	C30/37, f_{cd} = 20.00	top 4Ø20, bottom 3Ø20 (A_s = 2199 mm²)	Ø8 @ 200	36

SECTION	B × H [MM]	CONCRETE [MPA]	LONGITUDINAL BARS	LINKS	MEMBERS
Beam Y	300 × 500	C30/37, f_{cd} = 20.00	top 4Ø20, bottom 3Ø16 (A_s = 1860 mm²)	Ø8 @ 200	32

12 supports, 12 of them fully fixed. Member local axes: x from start to end node; for beams y is horizontal and z points up; for columns z is global X.

3 Actions

CASE	NAME	TYPE	LOADS	ΣF_X [KN]	ΣF_Y [KN]	ΣF_Z [KN]
G1	Self weight of frame and slabs	permanent + self weight	240	0	0	-6 540.6
G2	Finishes and facade	permanent	270	0	0	-2 232.0
Q	Imposed load, all floors	variable, Cat. B, office, ψ = 0.7, 0.5, 0.3	180	0	0	-1 782.0
S	Snow on the roof	variable, Snow, H <= 1000 m, ψ = 0.5, 0.2, 0	60	0	0	-198.0
W+X	Wind in +X	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	12	112.9	0	0
W-X	Wind in -X	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	12	-112.9	0	0
W+Y	Wind in +Y	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	16	0	184.7	0
W-Y	Wind in -Y	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	16	0	-184.7	0

4 Combinations of actions

74 ULS, 37 characteristic, 12 frequent and 2 quasi-permanent combinations, all checked.

5 Analysis results

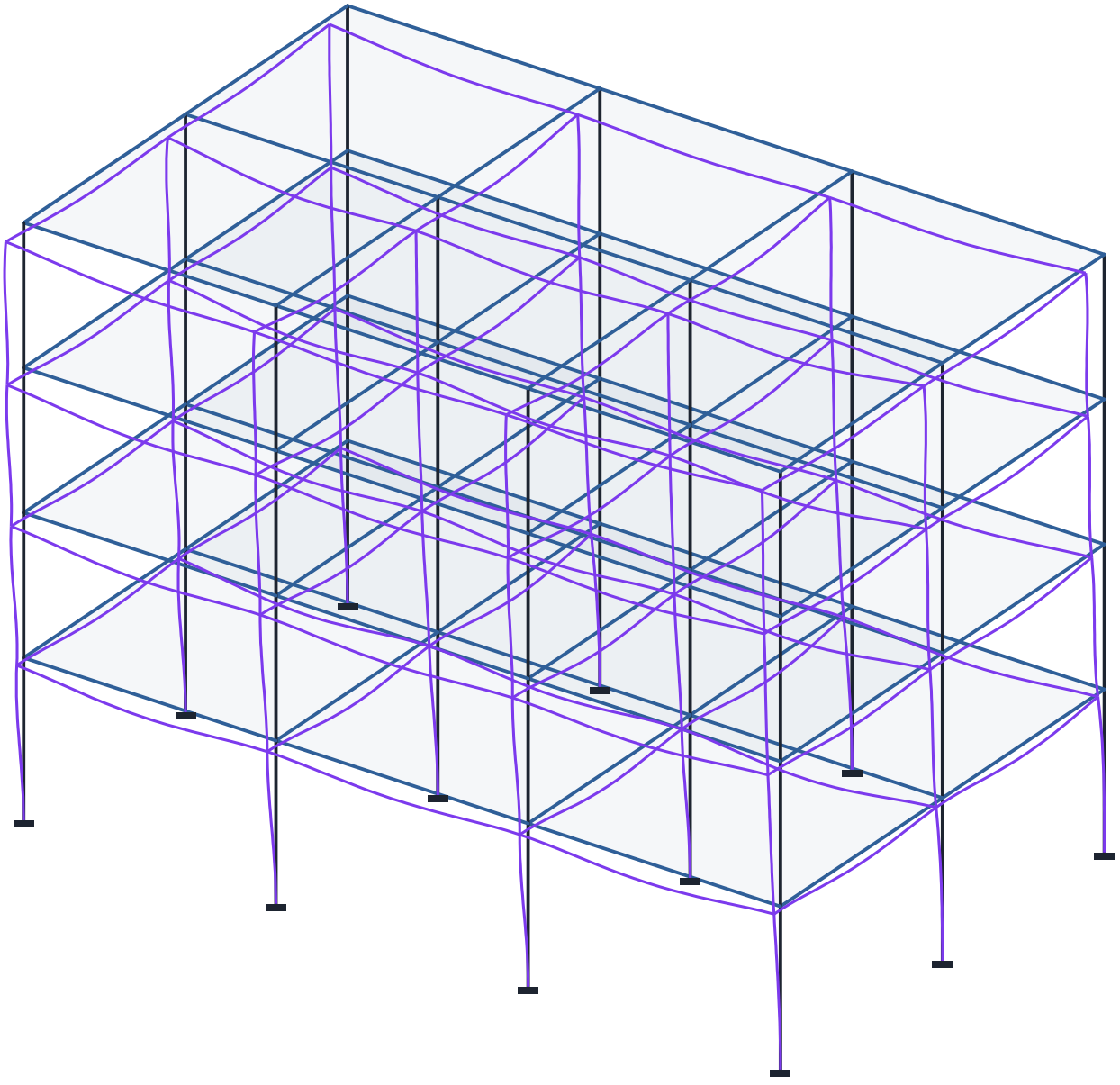
Equilibrium self-check: the largest residual between applied loads and reactions over all load cases is 2.5e-10 kN (or kNm).

Support reactions, ULS envelope (min … max)

NODE	R_X [KN]	R_Y [KN]	R_Z [KN]	M_X [KNM]	M_Y [KNM]	M_Z [KNM]
A1-L0	-3.2 … 25.6	-11.1 … 29.6	428.3 … 754.4	-54.7 … 32.4	-15.2 … 42.4	-0.7 … 0.6
B1-L0	-15.5 … 13.8	-15.3 … 39.8	728.3 … 1291.8	-72.6 … 43.2	-30.0 … 27.9	-0.5 … 0.5
C1-L0	-13.8 … 15.5	-15.3 … 39.8	728.3 … 1291.8	-72.6 … 43.2	-27.9 … 30.0	-0.5 … 0.5
D1-L0	-25.6 … 3.2	-11.1 … 29.6	428.3 … 754.4	-54.7 … 32.4	-42.4 … 15.2	-0.6 … 0.7
A2-L0	-3.7 … 33.7	-23.2 … 23.2	725.6 … 1250.3	-47.0 … 47.0	-18.2 … 54.5	-0.3 … 0.3
B2-L0	-19.0 … 16.8	-30.5 … 30.5	1186.4 … 2086.1	-61.3 … 61.3	-36.6 … 34.0	-0.3 … 0.3

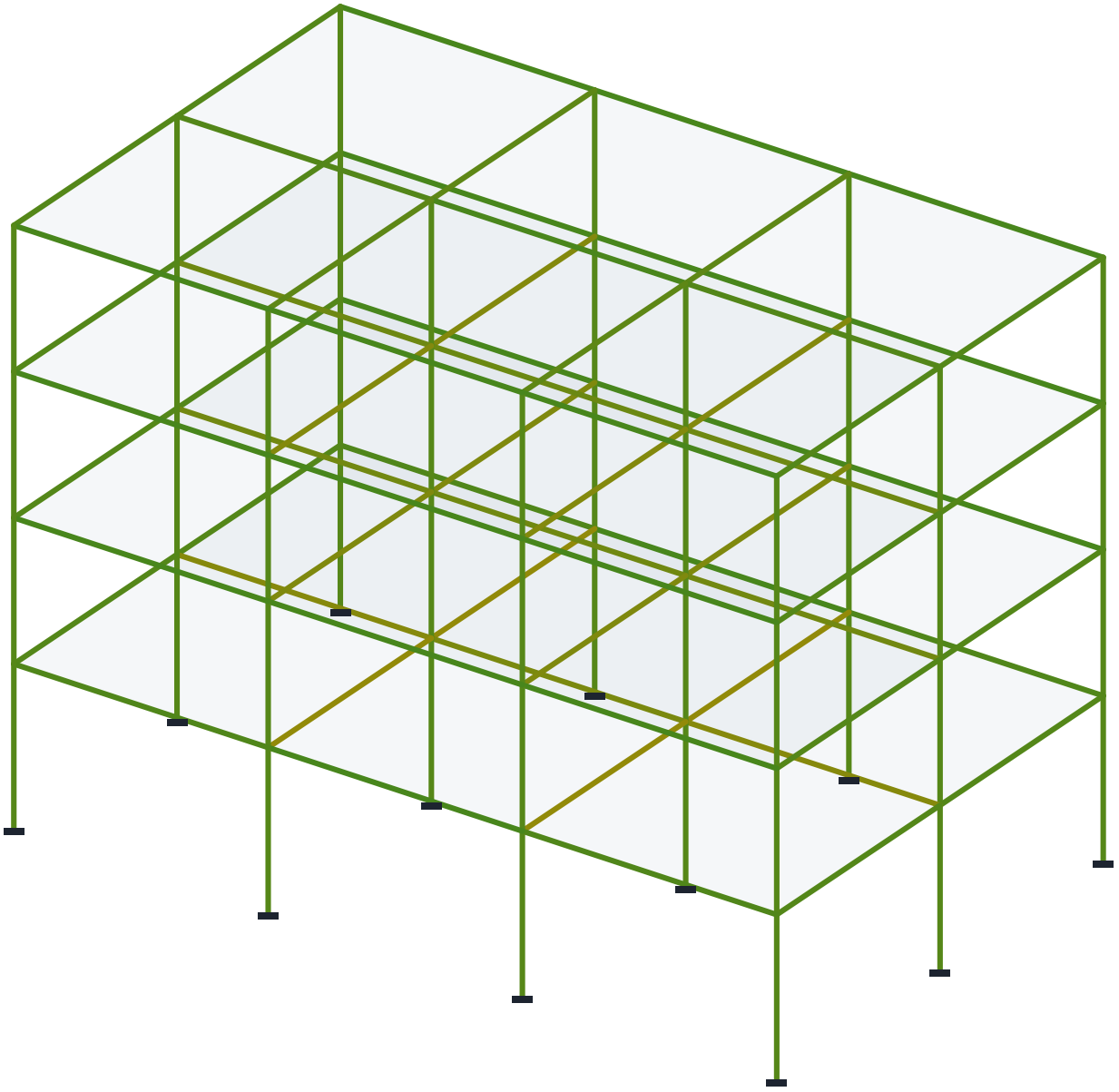
NODE	R_X [KN]	R_Y [KN]	R_Z [KN]	M_X [KNM]	M_Y [KNM]	M_Z [KNM]
C2-L0	-16.8 … 19.0	-30.5 … 30.5	1186.4 … 2086.1	-61.3 … 61.3	-34.0 … 36.6	-0.3 … 0.3
D2-L0	-33.7 … 3.7	-23.2 … 23.2	725.6 … 1250.3	-47.0 … 47.0	-54.5 … 18.2	-0.3 … 0.3
A3-L0	-3.2 … 25.6	-29.6 … 11.1	428.3 … 754.4	-32.4 … 54.7	-15.2 … 42.4	-0.6 … 0.7
B3-L0	-15.5 … 13.8	-39.8 … 15.3	728.3 … 1291.8	-43.2 … 72.6	-30.0 … 27.9	-0.5 … 0.5
C3-L0	-13.8 … 15.5	-39.8 … 15.3	728.3 … 1291.8	-43.2 … 72.6	-27.9 … 30.0	-0.5 … 0.5
D3-L0	-25.6 … 3.2	-29.6 … 11.1	428.3 … 754.4	-32.4 … 54.7	-42.4 … 15.2	-0.7 … 0.6

Deflected shape, CHAR035
max. displacement 5.0 mm, drawn ×190



Short-term elastic deflections, gross sections.

Largest utilisation per member
green: low, amber: close to 1, red: fails



6 Summary of design checks

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-B1-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-C1-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-B2-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-C2-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BX-A2-L1	beam	Beam X	Bending with axial force, both axes	0.7945	Pass
BX-C2-L1	beam	Beam X	Bending with axial force, both axes	0.7945	Pass
BY-C1-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-C2-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-B1-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-B2-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-B1-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-C2-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-C1-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-B2-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BX-B2-L1	beam	Beam X	Bending with axial force, both axes	0.7508	Pass
BX-A2-L2	beam	Beam X	Bending with axial force, both axes	0.7204	Pass
BX-C2-L2	beam	Beam X	Bending with axial force, both axes	0.7204	Pass
BX-B2-L2	beam	Beam X	Bending with axial force, both axes	0.7116	Pass
BX-A2-L3	beam	Beam X	Bending with axial force, both axes	0.7038	Pass
BX-C2-L3	beam	Beam X	Bending with axial force, both axes	0.7038	Pass
BX-B2-L3	beam	Beam X	Bending with axial force, both axes	0.6926	Pass
BY-C1-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-B1-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-B2-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-C2-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
C-A1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
C-B2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
BY-A1-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-D2-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A1-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A1-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A1-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BX-A2-L4	beam	Beam X	Bending with axial force, both axes	0.5828	Pass
BX-C2-L4	beam	Beam X	Bending with axial force, both axes	0.5828	Pass
BX-A3-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-C3-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-C1-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-A1-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-B2-L4	beam	Beam X	Bending with axial force, both axes	0.5403	Pass
BX-B1-L1	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L1	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BX-A1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass

7 Design checks in detail

Shown for every member that fails and the 12 most utilised members. Run with the full option for every member; all members appear in the summary above.

Member BY-B1-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 2.475 m from node i; combination ULS028: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.124	**-0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS019: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C1-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 2.475 m from node i; combination ULS028: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.124	**0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 2.750 m from node i; combination ULS011: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B2-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 3.025 m from node i; combination ULS036: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.124	**-0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 3.025 m from node i; combination ULS019: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366

Pass

Governing at x = 3.025 m from node i; combination ULS036: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.124	**0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS011: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BX-A2-L1 · Beam X · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7945 — Pass

Governing at x = 6.000 m from node i; combination ULS010: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+X

N_{Ed}	axial force (compression +) = -15.03	**-15.03** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = -199.5	**-199.5** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -5.242e-15	**-5.242e-15** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**-251.1** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**108.2** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 165 000·20.00 + 2199.1·434.8	**4 256.1** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.795^{1.00} + 0^{1.00}]^{1/1.00}$	**0.7945**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6323 — Pass

Governing at x = 6.000 m from node i; combination ULS010: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+X

V_{Ed}	design shear force = 156.1	**156.1** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 502.0)$	**0.0083**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -15030.0/165\,000$	**-0.0911** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00834 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0911] \cdot 300.0 \cdot 502.0$	**84.18** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 502.0$	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 451.8 \cdot 434.8 \cdot 2.500$	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 451.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 156.1/246.8$	**0.6323**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0076

Pass

Governing at x = 0.000 m from node i; combination ULS067: 1.00·G1 + 1.00·G2 + 1.50·W-Y

V_{Ed}	design shear force = 0.862	**0.8624** kN	
d	effective depth to the top (tension) reinforcement	**229.3** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/229.3)}$	**1.934**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(550.0 \cdot 229.3)$	**0.0075**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -13492.0/165\,000$	**-0.0818** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.934^{1.5} \cdot 30.00^{0.5}$	**0.5155** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.934 \cdot (100 \cdot 0.00747 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0818] \cdot 550.0 \cdot 229.3$	**80.99** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 229.3$	**206.4** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 206.4 \cdot 434.8 \cdot 2.500$	**112.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 550.0 \cdot 206.4 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**413.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**112.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.862/112.8$	**0.0076**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3332

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**2 199.1** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1805 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1394

Pass

Governing at x = 3.000 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**6 109.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 694.5** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**62.96** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5702**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**3.345** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 3.345/24.00	**0.1394**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BX-C2-L1 · Beam X · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7945 — Pass

Governing at x = 0.000 m from node i; combination ULS018: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-X

N_{Ed}	axial force (compression +) = -15.03	**-15.03** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = -199.5	**-199.5** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -6.361e-15	**-6.361e-15** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**-251.1** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**108.2** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 165 000·20.00 + 2199.1·434.8	**4 256.1** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.795^{1.00} + 0^{1.00}]^{1/1.00}$	**0.7945**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6323 Pass

Governing at x = 0.000 m from node i; combination ULS018: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-X

V_{Ed}	design shear force = 156.1	**156.1** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 502.0)$	**0.0083**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -15030.0/165\,000$	**-0.0911** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00834 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0911] \cdot 300.0 \cdot 502.0$	**84.18** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 502.0$	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 451.8 \cdot 434.8 \cdot 2.500$	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 451.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 156.1/246.8$	**0.6323**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0076 — Pass

Governing at x = 3.250 m from node i; combination ULS030: 1.35·G1 + 1.35·G2 + 1.50·W-Y

V_{Ed}	design shear force = 0.862	**0.8624** kN	
d	effective depth to the top (tension) reinforcement	**229.3** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/229.3)}$	**1.934**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(550.0 \cdot 229.3)$	**0.0075**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -18211.0/165\,000$	**-0.1104** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.934^{1.5} \cdot 30.00^{0.5}$	**0.5155** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.934 \cdot (100 \cdot 0.00747 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.11] \cdot 550.0 \cdot 229.3$	**80.44** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 229.3$	**206.4** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 206.4 \cdot 434.8 \cdot 2.500$	**112.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 550.0 \cdot 206.4 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**413.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**112.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.862/112.8$	**0.0076**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3332 — Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**2 199.1** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1805 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ \text{mm})$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ \text{mm})$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1394

Pass

Governing at x = 3.000 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**6 109.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 694.5** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**62.96** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5702**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**3.345** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 3.345/24.00	**0.1394**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C1-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.087	**-0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·452.0	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 123.7/222.3	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 3.025 m from node i; combination ULS031: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.087	**-0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 123.7/222.3$	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 0.000 m from node i; combination ULS023: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B1-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.087	**0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·452.0	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 123.7/222.3	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 0.000 m from node i; combination ULS031: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B2-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.087	**0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 123.7/222.3$	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 2.750 m from node i; combination ULS023: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd} / (\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00 / (2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, $\beta = 0.5$, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B1-L2 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7683 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = 2.267	**2.267** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.13	**87.13** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.0789	**0.0789** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**113.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**94.09** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 5.952e-04	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.767^{1.00} + 8.388e\text{-}04^{1.00}]^{1/1.00}$	**0.7683**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5799 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 128.9	**128.9** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 13833.8/150 000	**0.0922** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0922] \cdot 300.0 \cdot 452.0$	**83.96** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·452.0	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 128.9/222.3	**0.5799**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0067

Pass

Governing at x = 2.750 m from node i; combination ULS007: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.751	**0.7505** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ = 829.4/(500.0·226.7)	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 2509.0/150 000	**0.0167** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ 0.15·0.0167]·500.0·226.7	**74.15** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·226.7	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ = 0.503·204.1·434.8·2.500	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ = 1.0·500.0·204.1·0.528·20.00/(2.500 + 0.4)	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} = 0.751/111.5	**0.0067**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ = 0.04·150 000	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1275

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5364**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.804** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.804/22.00	**0.1275**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L2 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7683 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = 2.267	**2.267** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.13	**87.13** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.0789	**-0.0789** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**113.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-94.09** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 5.952e-04	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.767^{1.00} + 8.388e\text{-}04^{1.00}]^{1/1.00}$	**0.7683**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5799 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 128.9	**128.9** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 13833.8/150 000	**0.0922** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0922] \cdot 300.0 \cdot 452.0$	**83.96** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 128.9/222.3$	**0.5799**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0067

Pass

Governing at x = 3.025 m from node i; combination ULS015: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.751	**0.7505** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 2509.0/150 000	**0.0167** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0167] \cdot 500.0 \cdot 226.7$	**74.15** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} = 0.751/111.5	**0.0067**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ = 0.04·150 000	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020 — Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1275 — Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5364**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.804** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.804/22.00	**0.1275**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

8 Assumptions and limitations

- Linear-elastic first-order analysis with gross sections; slender columns get second-order moments by the nominal curvature method on the isolated member (5.8.8) with the effective lengths set in the model.
- The generator assumes a braced frame (effective length = storey height). For an unbraced sway frame, increase the column effective length factors.
- Slabs are not designed and not modelled as plates; there is no rigid-diaphragm constraint. Wind is applied at the windward facade nodes.
- Biaxial bending by EN 1992-1-1 Eq. 5.39 using exact uniaxial resistances; shear checked in each plane separately; torsion shown where $T_{Ed} \geq 0.1\cdot T_{Rd,c}$.
- Reinforcement uniform along each member; anchorage, laps, crack widths and punching are not checked.
- A licensed engineer remains responsible for the model, its inputs and the use of these results.

9 Software verification

This build ran its verification suite when producing this report: 35 benchmarks with 169 comparisons, **all passed**.

Generated by Tragwerk 0.2.0 on 2026-10-21; analysis time 3.39 s.